

< Papers  arxiv:2507.13334 

A Survey of Context Engineering for Large Language Models

Published on Jul 17, 2025

★ Submitted by  [Lingrui Mei](#) on Jul 18, 2025

Authors:  [Lingrui Mei](#),  [Jiayu Yao](#),  [Yuyao Ge](#),
[Yiwei Wang](#),  [Baolong Bi](#), [Yujun Cai](#),  [Jiazhi Liu](#),
 [Mingyu Li](#),  [Zhong-Zhi Li](#),  [Duzhen Zhang](#),
 [Chenlin Zhou](#),  [Jiayi Mao](#), [Tianze Xia](#), [Jiafeng Guo](#),
[Shenghua Liu](#)

Abstract

Context Engineering systematically optimizes information payloads for Large Language Models, addressing gaps in generating sophisticated, long-form outputs.

🔮 AI-generated summary

The performance of [Large Language Models](#) (LLMs) is fundamentally determined by the contextual information provided during inference. This survey introduces [Context Engineering](#), a formal discipline that transcends simple prompt design to encompass the systematic optimization of information payloads for LLMs. We present a comprehensive taxonomy decomposing [Context Engineering](#) into its foundational components and the sophisticated implementations that integrate them into intelligent systems. We first examine the foundational components: [context retrieval](#) and [generation](#), [context processing](#) and [context management](#). We then explore how these components are

▲ Upvote **260**



 Models citing this paper 0

No model linking this paper

Cite arxiv.org/abs/2507.13334 in a model README.md to link it from this page.

 Datasets citing this paper 0

No dataset linking this paper

Cite arxiv.org/abs/2507.13334 in a dataset README.md to link it from this page.

 Spaces citing this paper 0

No Space linking this paper

Cite arxiv.org/abs/2507.13334 in a Space README.md to link it from this page.

 Collections including this paper 62

 Ask HuggingChat about this Paper



architecturally integrated to create sophisticated system implementations: retrieval-augmented generation (RAG), memory systems and tool-integrated reasoning, and multi-agent systems. Through this systematic analysis of over 1300 research papers, our survey not only establishes a technical roadmap for the field but also reveals a critical research gap: a fundamental asymmetry exists between model capabilities. While current models, augmented by advanced context engineering, demonstrate remarkable proficiency in understanding complex contexts, they exhibit pronounced limitations in generating equally sophisticated, long-form outputs. Addressing this gap is a defining priority for future research. Ultimately, this survey provides a unified framework for both researchers and engineers advancing context-aware AI.

[↗ View arXiv page](#)

[↗ View PDF](#)

[🐙 GitHub](#) ★ 2.85k

[🗂 Add to collection](#)

 Community

 **Chevalier**

[Paper author](#)

[Paper submitter](#)

Jul 17, 2025

⋮

Ongoing work; 165 pages, 1401 citations.

[See translation](#)

 3 replies

 3



deleted Jul 18, 2025

This comment has been hidden



> Expand 2 replies

Reply in thread

- Agent & RL**  Collection
55 items • Updated Nov 27, 2025 •  21
- Papers to Read**  Collection
208 items • Updated Aug 24, 2025 •  10
- interesting stuff**  Collection
162 items • Updated 20 days ago •  6

[Browse 62 collections that include this paper](#)





 **TestregX** Jul 18, 2025



 2 replies · 

deleted Jul 18, 2025

This comment has been hidden



> Expand 1 reply

Reply in thread

deleted Jul 18, 2025

This comment has been hidden



 **panikov** Jul 18, 2025



wow, very cool

thanks for extensive taxonomy

[See translation](#)

 1 +

 Reply

 **Chevalier** [Paper author](#) [Paper submitter](#) Jul 20, 2025



I wrote the wrong email address in my paper. The correct email address should be: meilingrui25b@ict.ac.cn

[See translation](#)



 Reply

 **WVest** Jul 21, 2025



What an amazing job! Thank you for this piece of what will be historical cannon

[See translation](#)



🟡 **Safeswimming69** Jul 24, 2025 ⋮

Wows 166 pages, thanks for the tldr summary :)

[See translation](#)



↩ Reply

🌈 **SunSpace0923** Oct 25, 2025 ⋮

牛逼



↩ Reply

🟡 **avahal** Dec 23, 2025 ⋮

arXiv lens breakdown of this paper 📌

<https://arxivlens.com/PaperView/Details/a-survey-of-context-engineering-for-large-language-models-1297-2cab1382>

- Executive Summary
- Detailed Breakdown
- Practical Applications

[See translation](#)



↩ Reply

Edit

Preview

Start discussing about this paper



Upload images, audio, and videos by dragging in the text input, pasting, or [clicking here](#).





Comment

· [Sign up](#) or [log in](#) to comment



System theme

[TOS](#)

[Privacy](#)

[About](#)

[Careers](#)



[Models](#)

[Datasets](#)

[Spaces](#)

[Pricing](#)

[Docs](#)

